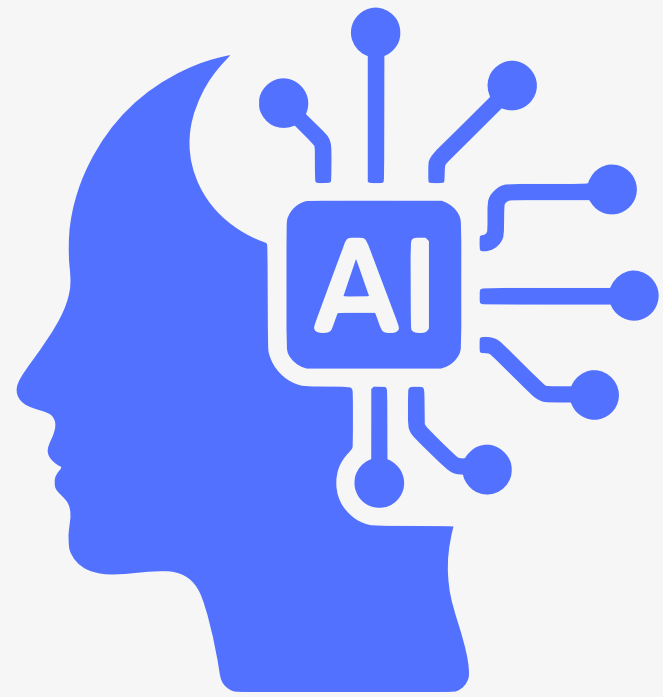




HYBRID SEARCH & RE-RANKING

FINDING INFORMATION IS EASY. FINDING THE RIGHT INFORMATION IS
WHAT MAKES A PRODUCTION RAG SYSTEM TRULY INTELLIGENT.





WHAT IS HYBRID SEARCH?

- Combines multiple search techniques
- Uses both keyword and semantic search
- Improves retrieval accuracy
- Reduces irrelevant results
- Industry standard for Enterprise RAG systems





WHAT IS BM25?

- Traditional keyword-based search algorithm
- Matches exact words in documents
- Fast and reliable
- Great for names, IDs, and technical terms
- Used by search engines like Elasticsearch





WHAT IS VECTOR SEARCH?

- Searches by meaning, not exact words
- Uses embeddings
- Understands semantic similarity
- Great for natural language queries
- Core technology behind modern AI search





BM25 VS VECTOR SEARCH

BM25

- Exact keyword matching & Fast
- Doesn't understand meaning
- Best for technical keywords

Vector Search

- Semantic search
- Understands context
- Handles paraphrased queries
- Better for AI applications





WHAT IS HYBRID SEARCH?

- Combines BM25 + Vector Search
- Retrieves better candidate documents
- Balances precision and semantic understanding
- Used by enterprise AI systems
- Improves RAG quality significantly





WHAT IS RE-RANKING?

- Reorders retrieved documents
- Selects the most relevant context
- Removes weaker matches
- Improves answer quality
- Happens after retrieval





WHAT IS COHERE RE-RANK?

- AI-powered re-ranking model
- Scores retrieved documents
- Sorts results by relevance
- Improves RAG accuracy
- Widely used in production AI systems

